

SpeechMate: An AI-Powered Communication and Speech Training Assistant with Accent-Aware Pronunciation Assessment and Deferred Multimodal Feedback

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Abstract—*Weak communication skills are a leading cause of graduate unemployment in Malaysia, with 64% of employers citing poor English and 53% citing weak communication and presentation skills. Existing AI speech-coaching applications leave three gaps unaddressed: they score pronunciation against native-English norms rather than fairly assessing accent, they deliver feedback in real time in a way that raises cognitive load, and none correlate gaze behaviour with vocal disfluency on one timeline. This paper presents SpeechMate, an AI-powered communication training platform combining automatic speech recognition, computer-vision-based behaviour analysis, and an LLM coach to address these gaps directly, rather than as incidental side effects of a more general feature set. The system uses accent-aware Goodness-of-Pronunciation scoring, a webcam-only gaze-disfluency fusion metric (“Gaze Tunneling”), and a deferred, post-session feedback design grounded in Cognitive Load Theory. The speech-tracking pipeline is implemented and operational; development is currently focused on improving pronunciation-tracking accuracy and feedback specificity, informed by early user testing, with the diagnostic reasoning behind each fix reported explicitly rather than presented as a solved problem. A systematic comparison against six current commercial products (Yoodli, Orai, Elqo, ELSA Speak, BoldVoice, Poised) confirms that none combine accent-fair assessment, deferred-only feedback, and fused gaze-disfluency correlation. The contribution reported here is therefore one of system architecture and implementation strategy: a multimodal design whose routing, scoring and feedback-timing decisions each trace to an identified gap, realised end to end on commodity hardware, together with a traceability matrix and a pre-registered protocol that fix in advance how the design is to be validated.*

Keywords—*Software systems and architecture, ICT in education, multimodal system design, speech recognition, accent-aware pronunciation assessment, deferred feedback, cognitive load theory*

I. INTRODUCTION

Communication competence is a determining factor in the employability of graduates in Malaysia. A national survey reported that 64% of employers cited poor English proficiency and 53% cited poor communication and presentation skills as the main reasons graduates failed interviews and remained unemployed [1], [2]. The barrier is clearly not subject knowledge but the ability to deliver it under pressure, and a growing category of AI speech guidance applications has emerged in response.

However, as this paper demonstrates through a systematic comparison of both the academic literature and current commercial products, existing tools consistently fall short in three specific ways. First, they treat non-native accent as an error to be corrected rather than a legitimate variation to be

fairly assessed. Second, they lean into real-time, in-session feedback as a selling point, without accounting for the cognitive cost this imposes on an already-anxious speaker. Third, none of the reviewed systems fuse gaze behaviour and vocal disfluency into a single correlated metric, even though both signals are frequently captured by the same platform. SpeechMate is proposed specifically to close these three gaps within one coherent, locally-deployable system.

A. Problem Statement and Research Gap

Four problems motivate the work. Students working across languages hit blocks when converting thought into formal English under pressure, and anxiety and stutters follow [3]. Commercial language-learning applications build vocabulary but keep learners on isolated words and rehearsed phrases rather than spontaneous talk [4]. Practice tools that display a script invite the user to read it while speaking, entrenching the habit this paper calls Gaze Tunneling and denying any practice at genuine audience contact [5]. And most trainers fire lights, warnings or timers while speech is still underway, fracturing working memory and compounding anxiety mid-sentence [6].

Surveying what is currently sold confirms these gaps persist. Orai, Speeko and Poised return quick verdicts on fillers, pace and energy, but listen only to audio and never reach phoneme level [7], [8]. Yoodli supports AI roleplay and live-meeting coaching, yet its pricing and positioning target enterprise sales teams rather than students [9]. Elqo is the nearest multimodal analogue, reporting voice, eye contact, facial expression and body language together [10] but reporting them side by side, never as one fused quantity over a shared timeline. ELSA Speak and BoldVoice do score individual phonemes, though explicitly in service of moving a speaker toward American English rather than judging their accent on its own terms [11], [12]. Notably, every one of these treats immediacy of feedback as a virtue; none defers it deliberately on cognitive-load grounds.

Taken together, this comparative analysis, developed further in Section III, identifies four specific gaps that this paper addresses: (i) the absence of accent-fair, rather than accent-corrective, pronunciation assessment; (ii) the absence of a deferred-feedback design informed by Cognitive Load Theory; (iii) the absence of a fused gaze-disfluency correlation metric; and (iv) the absence of a locally-deployable option in which session recordings need not leave the user’s machine.

B. Objectives and Contributions

This study pursues four objectives: to develop an AI-powered pre-presentation scripting assistant that lowers upfront cognitive load; to build a dialogue system evaluating spoken proficiency using suprasegmental features alongside segmental phonetic accuracy with accent-aware thresholds; to implement a multimodal fusion pipeline correlating webcam gaze coordinates against vocal stutter timestamps on

a unified session timeline; and to provide a deferred, post-session analytics dashboard that withholds scoring until after the session ends, consistent with Cognitive Load Theory.

The contributions of this paper are as follows.

- A dual-engine ASR routing design in which a fast general-purpose model serves simultaneously as transcriber and as the language-confidence router, reserving a heavier code-switch-specialised model for the inputs that actually require it, rather than paying that cost on every session.
- An accent-fair pronunciation-scoring formulation that separates deviation from reference pronunciation into documented regional variation and intelligibility-affecting error, rather than applying a single native-norm threshold.
- A fused gaze-disfluency metric computed as the correlation between gaze aversion and disfluency occurrence over a shared session timeline, rather than reporting the two as independent statistics.
- A deferred-feedback interaction design motivated explicitly by Cognitive Load Theory, in contrast to the real-time feedback paradigm adopted by every commercial system reviewed.
- A systematic comparison against six current commercial AI speech-coaching products, establishing that no existing system combines these properties, together with a pre-registered evaluation protocol specifying acceptance thresholds in advance of measurement.

C. Outline of the Paper

The remainder of this paper is organised as follows. Section II describes the system architecture, development methodology, and pronunciation-assessment pipeline. Section III reviews related academic and commercial work and presents the comparative analysis that motivates this paper's contribution. Section IV reports current implementation status, preliminary user feedback, and the diagnostic steps taken to address an identified pronunciation-accuracy limitation. Section V concludes the paper and outlines future work.

II. METHODOLOGY

This study adopts a design-based development approach, combining iterative system development with early, informal user validation, so that technical design choices remain grounded in how real users actually experience the practice session rather than in specification alone.

A. Development Approach

Development followed Agile across six sprints, each carrying one capability: authentication, profiles and goal tracking; scenario-based conversational AI spanning casual, interview and presentation modes; speech analysis covering transcription, fluency and disfluency; computer vision for eye contact, posture and expression; analytics fusing the speech and vision outputs into one deferred report; and testing, which corresponds to the protocol of Section IV.

Agile was selected because AI-dependent components can only be assessed empirically once real data is available, allowing sprint scope to be adjusted rather than fixed before any model has been evaluated on real users.

B. System Architecture

Fig. 1 shows the layered design. Capture happens wholly in the browser: MediaPipe Tasks Vision (FaceLandmarker, PoseLandmarker) tracks face mesh and pose on the client while the Web Audio API monitors microphone level live. The interface is a Next.js 16 application (App Router, React 19, TypeScript) styled with Tailwind CSS v4, holding client state in Zustand and drawing scores through Recharts. Behind it, a FastAPI service on Python 3.13 with JWT authentication exposes routes for practice, speech, analysis, vision, coaching, reporting and session websockets. Results persist through async SQLAlchemy over SQLite with Alembic migrations, in a schema where each session fans out into speech-analysis, vision-analysis and feedback records plus per-metric history rows that feed the trend charts.

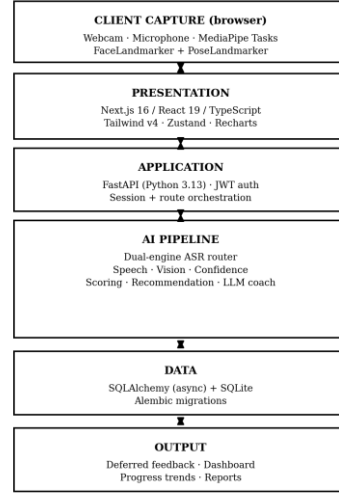


Fig. 1. SpeechMate layered system architecture.

C. Dual-Engine ASR Routing for Code-Switched Speech

How to transcribe speech that mixes English with Bahasa Malaysia is a central implementation question. Instead of trusting one general multilingual model, the system routes between two. A faster-whisper checkpoint (small, CPU int8, word-level timestamps) transcribes first and simultaneously supplies the language-confidence score on which routing turns. When that score does not indicate confident English, the audio passes again through Mesolitica's wav2vec2-xls-r-300m-mixed checkpoint [24], trained on Malay, Singlish and Mandarin-mixed speech and loaded through PyTorch and transformers directly rather than via a wrapper toolkit. Its card reports 0.132 WER and 0.048 CER on the authors' own evaluation; those figures characterise the checkpoint chosen and are not results of this system.

This routing is the concrete mechanism behind the code-switching support claimed in Section I, and reflects a deliberate trade-off: rather than accepting the degraded accuracy an English-centric model shows on Manglish, or paying the cost of a large multilingual model on every session, a fast general model serves as both transcriber and router, reserving the specialised model for inputs that require it. The specialised checkpoint is approximately 1.2 GB and is therefore fetched on first non-English transcription rather than bundled with the application.

A complementary component, a documented lexicon of roughly 190 Bahasa Malaysia and Manglish terms, is used to estimate the English-to-Malay ratio within a transcript and flag code-switching. This is an explicit word-list heuristic rather than a trained language-identification model, and is

treated as such throughout: it is adequate for characterising the degree of code-switching in a session for feedback purposes, but is not presented as a validated language-identification method, and its coverage is bounded by the lexicon’s size.

D. Accent-Fair Pronunciation Assessment

Pronunciation is scored by Goodness-of-Pronunciation forced alignment, with acceptance thresholds loosened relative to standard practice — a decision following evidence that alignment error predicts pronunciation score only weakly, and most strongly for plosives and fricatives [13]. Rather than loosening uniformly, deviations are sorted into two kinds: those matching documented Malaysian-English phonology, which count as variation and go unpenalised, and those liable to cost a listener comprehension, which are surfaced. Fig. 2 sketches the distinction. Fairness therefore becomes a stated, inspectable property of the scoring rule instead of a hidden threshold tweak, and it is what separates this system from accent-reduction tools.

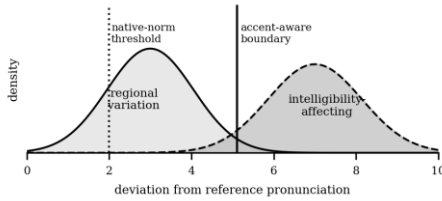


Fig. 2. Accent-fair scoring. A native-norm threshold penalises regional variation; the accent-aware boundary separates variation from intelligibility-affecting error. Illustrative, not measured.

Session recordings are handled under an ephemeral-processing design: raw audio and video are used only to extract the derived features described above, and are retained only for a short, configurable review window before deletion, consistent with the data-minimisation principle underpinning Malaysia’s Personal Data Protection Act 2010. This is a deliberate design choice rather than an incidental one, since anxious users – the population most explicitly targeted by this work – are also plausibly the most sensitive to the idea of a recording of themselves stumbling through practice being retained indefinitely.

E. Multimodal Fusion: The Gaze Tunneling Metric

Fillers, repetitions and prolonged pauses are stamped onto the transcript timeline, and per-frame gaze-aversion state is stamped onto the same one. Gaze Tunneling is then the Pearson correlation between aversion and disfluency across that timeline. Fig. 3 conveys the intuition. This is what makes the measure fused rather than parallel: the system does not merely place an eye-contact percentage beside a disfluency count, it asks whether the two coincide, supporting feedback such as “you tend to look away exactly when you stumble” instead of two numbers a user must reconcile unaided. No reviewed commercial system reports such a correlation.

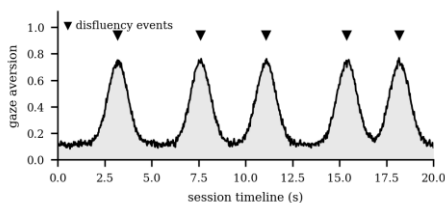


Fig. 3. Schematic of the Gaze Tunneling metric: gaze-aversion trace with disfluency events marked on a shared timeline. Co-

occurrence is what the correlation quantifies. Trace is illustrative, not measured.

F. Implementation Status

The system is functional end-to-end for its core loop: a user records a practice session, the multimodal pipeline analyses it, and the interface returns a communication score, deferred feedback, and updated progress trends. All AI components operate on pretrained inference only; no model fine-tuning or training was performed in this work, which bounds what can be claimed about accuracy on Malaysian-accented speech and is treated explicitly as a limitation in Section IV.E. Current development effort is concentrated on improving pronunciation-tracking accuracy and the specificity of generated coaching feedback, both discussed in Section IV.

III. LITERATURE REVIEW

A. Cognitive Load Theory and Deferred Feedback

Sweller’s Cognitive Load Theory holds that working memory has a strictly limited capacity, and that when intrinsic and extraneous processing demands exceed this threshold, cognitive overload occurs and verbal performance collapses [14]. Carless, and Butler and Winne, similarly emphasise that feedback must be timed and structured to support learning rather than provoke defensive or anxious responses [6], [15]. Taken together, these findings motivate SpeechMate’s core design decision to withhold all scoring and feedback until after a practice session ends, in direct contrast to the real-time feedback paradigm adopted by nearly every reviewed commercial competitor.

B. Automatic Speech Recognition and Accent Robustness

Contemporary ASR relies on deep models trained over large speech corpora to recover words, phonemes and sentence structure at high accuracy [16]. Forced-alignment pronunciation scoring, however, carries measurement noise of its own: Mathad et al. observed boundary errors of 30–60 ms yet found alignment error a weak overall predictor of the resulting score, its influence concentrated in plosives and fricatives [13]. Strict thresholds would therefore punish noise as though it were mispronunciation, which argues for the relaxed setting adopted here.

C. AI-Based Speaking Assistants and Conversational Coaching

Qin et al. propose an LLM-driven speaking assistant that provides real-time text suggestions as speaking references for non-native speakers [21]. The approach improves logical structuring of a speaker’s response in real time, but relies on typed, on-screen prompts during active speech, which increases cognitive load and risks language hallucination. This motivates SpeechMate’s use of passive, silent multimodal tracking during practice, with all coaching output deferred to session end rather than injected into the conversation itself.

D. Stuttering and Disfluency Detection

Kourkounakis et al. propose FluentNet, combining Squeeze-and-Excitation residual convolutional networks, bidirectional LSTMs, and an attention mechanism for end-to-end stuttered-speech detection, achieving high accuracy on the UCLASS corpus [17]. Its principal limitation is reliance on data such as LibriStutter, which does not fully

capture real psychological speech blocks, motivating SpeechMate's planned use of locally-collected and augmented data rather than purely synthetic corpora when the disfluency-detection component is trained.

E. Presentation Behaviour and Gaze Tracking

Du, Zhang and Lan present a real-time wearable system that analyses a speaker's gaze distribution across audience and non-audience regions, increasing eye-contact duration by 62.5% on average in a user study [18]. The approach depends on specialised wearable hardware and delivers real-time voice prompts during active speech. SpeechMate re-implements the same tri-zone gaze-distribution concept using only a standard webcam, logging gaze data silently for correlation with vocal disfluency events rather than interrupting the user in real time – trading a degree of tracking precision for zero additional hardware cost and full alignment with the deferred-feedback design established in Section III.A.

F. Accent Bias in Automatic Speech Recognition

Alignment noise is not the only source of error. A substantial literature documents that commercial ASR degrades on speakers from under-represented dialect groups relative to the standard-accent speech these systems are trained and benchmarked on, with large word-error-rate gaps across providers [22]. That work concerns dialect rather than second-language accent, so it transfers by analogy rather than directly; the analogy is nonetheless instructive. Accent-related error is a predictable consequence of training and evaluation practice, not a stray bug, which is why this work treats fairness as something to design for rather than calibrate away afterwards. Malaysian English and English-Bahasa Melayu switching are, on the same reasoning, improbably well covered by general-purpose commercial systems, hence the emphasis here on local data collection rather than parameter adjustment alone.

G. Comparative Analysis with Existing Commercial Systems

Fig. 4 summarises capability coverage across the reviewed systems. Six current AI-powered speech-coaching products were identified through a systematic review of the commercial landscape as of 2026.

Deferred-only feedback	–	–	●	–	●
Accent-fair pronunciation	–	–	–	●	●
Fused gaze-disfluency	–	–	●	–	●
Code-switch support	–	●	–	–	●
Local deployment / on-device data	–	●	–	–	●
	Oral / Speeko	Yoodli	Elqo	ELSA / BoldVoice	Speech-Mate

Fig. 4. Capability coverage across reviewed systems. ● full support ○ partial – absent

Within the AI decision-support literature, Latif et al. and Saleem et al. target predictive analytics and automated grading, leaving spoken delivery untouched [19], [20]; Zulmuksah and Mahamad likewise build an adaptive assessment and recommendation system validated through surveys, User Acceptance Testing and expert interviews [23]. That mixed-methods pairing of model metrics with stakeholder judgement shapes the protocol in Section IV.D, although the present domain and technical contribution

differ. Turning to commercial products: Orai, Speeko and Poised omit phoneme-level assessment altogether [7], [8]; Yoodli is capable but priced for enterprise sales teams [9]; Elqo comes closest to the multimodal ambition here yet keeps gaze and disfluency as unrelated readouts [10]; and ELSA Speak and BoldVoice, despite offering the richest pronunciation feedback commercially available, aim at reducing accent toward native norms rather than judging it fairly [11], [12]. None combines the four capabilities listed in Section I.A.

Data handling is a further point of differentiation. All six reviewed products store session data in the cloud; none advertise an ephemeral-processing mode. For a system targeting users with speaking anxiety, whether a recording of a difficult session persists plausibly affects whether such users practise honestly at all, which is why SpeechMate's ephemeral-session design is treated as a design requirement rather than a deployment detail.

H. Research Gaps

Three gaps recur across both academic and commercial work. First, real-time speaking-assistant tools intervene during active speech in ways that raise cognitive load rather than reduce it, and market this as a strength. Second, accent-focused tools are designed to correct accent toward a native-English norm rather than fairly distinguish accent from error, leaving Malaysian-accented and code-switched English users underserved. Third, no reviewed system, academic or commercial, fuses gaze and vocal-disfluency signals into a single correlated metric; the closest comparator (Elqo) reports these as separate parallel scores. SpeechMate is positioned to address all three gaps simultaneously.

IV. RESULTS AND DISCUSSION

As the system is under active development, this section reports current implementation status and preliminary, informal user feedback rather than final quantitative results, which will follow structured evaluation and be reported in a subsequent publication once model training, threshold calibration, and full User Acceptance Testing are complete.

A. Implementation Status

The core loop is functional end-to-end: a session is recorded in the browser, routed through the dual-engine ASR stage, analysed across speech and vision channels, fused into the Gaze Tunneling correlation and a communication score, and returned to the user as deferred feedback with updated progress trends. All components run on pretrained inference; no fine-tuning was performed. Development effort is concentrated on two areas identified through early, informal user testing: the specificity of AI-generated coaching feedback, and the accuracy of pronunciation tracking. Both are treated in this work as design problems with identifiable causes, rather than issues to be resolved through blind parameter tuning.

B. Diagnosing the Pronunciation-Accuracy Limitation

Users repeatedly described pronunciation scoring as inconsistent. Rather than tune parameters blindly, three causes are being examined in order of expected impact. An acoustic model not tuned locally can shift phoneme boundaries, and that drift propagates straight into GOP scores, matching the mechanism Mathad et al. describe [13]. Standard GOP references are built on native-accent pronunciation, so systematic Malaysian-English variation is

liable to be recorded as error. And thresholds taken from published defaults were never calibrated against locally collected audio.

The planned mitigation reflects the accent-aware positioning argued for in Section III: rather than uniformly relaxing thresholds, subsequent work will separate scoring output into two explicit categories – likely accent variation versus likely intelligibility-affecting error – so that accent-fairness becomes a visible, user-facing property of the system rather than an invisible backend adjustment. This distinction is expected to directly address the specific complaint raised in early testing, since a blended score currently gives no indication of which category a flagged phoneme falls into.

C. Improving Feedback Quality

Feedback generation is simultaneously being rebuilt around a fixed schema restricted to features the system already computes: filler count, words per minute, pause pattern, phoneme-level GOP, gaze-tunneling percentage and required to cite specific moments rather than generate open prose. The motivation is the recurring finding that unanchored coaching advice is judged less useful than advice tied to concrete points in a performance [21]. Fig. 5 shows why this anchoring is available: transcript, gaze and pronunciation events share a timeline before feedback is composed, so any statement can in principle be traced to a moment in the recording.

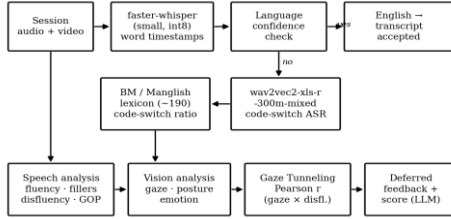


Fig. 5. Timestamp-aligned data-flow pipeline from capture to deferred feedback.

Two enhancements follow directly from this alignment and are detailed in Section V.B: exposing each flagged event as a short replayable clip alongside its explanation, and adapting scenario difficulty to a rolling baseline of the user's own recent sessions.

D. Planned Evaluation Protocol

The full evaluation protocol planned for the completed system follows the metrics established in Table I, combining objective measures with Likert-scale User Acceptance Testing of feedback specificity, actionability, and perceived accuracy. Each metric is paired with an explicit success criterion, summarised in Fig. 6, so that the eventual results can be assessed against a predetermined standard rather than interpreted post hoc. Setting thresholds before evaluation prevents acceptance criteria from being retrofitted to whatever results emerge.

TABLE I. PLANNED EVALUATION PROTOCOL

Component	Metric	Success criterion
Speech recognition	WER (English vs. code-switched)	Measurable WER reduction vs. untuned Whisper baseline
Pronunciation	Correlation with human raters; precision/recall	Correlation ≥ 0.7
Disfluency detection	Precision, Recall, F1; false-positive rate	F1 ≥ 0.75 ; low false positives on natural pauses
Gaze / eye contact	Accuracy vs. manually coded video	Accuracy ≥ 0.8

Feedback quality	Likert: specificity, actionability, accuracy	Significant gain after prompt restructuring
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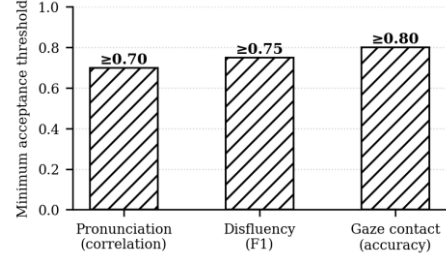


Fig. 6. Pre-registered minimum acceptance thresholds for the three quantitative components. Targets are set in advance of evaluation; no results are reported yet.

Table II traces each research objective to the component that implements it and the metric that will test it, so that no objective is left without an implementing mechanism or a means of falsification.

TABLE II. OBJECTIVE-MECHANISM-EVALUATION TRACEABILITY

Objective	Implementing mechanism	Validating metric	Status
O1 – lower upfront cognitive load	Pre-presentation scripting assistant	Likert: preparedness, perceived load	Implemented
O2 – suprasegmental + segmental scoring	GOP with accent-aware thresholds; fluency and pace analysis	Correlation with human raters; WPM and pause accuracy	Implemented; accuracy under revision
O3 – fuse gaze with disfluency	Shared-timeline alignment; Pearson correlation	Detection accuracy vs. human-coded video	Implemented
O4 – withhold feedback in session	Deferred post-session report; no in-session alerts	Likert: anxiety, interruption; within-subject comparison	Implemented; premise untested
Cross-cutting – code-switched input	Dual-engine ASR routing; ~190-term lexicon	WER on code-switched vs. English-only clips	Implemented; unevaluated

V. CONCLUSION

This paper introduced SpeechMate, an AI-powered communication and speech training platform designed to address three specific, evidence-based gaps in existing AI speech-coaching tools: the absence of accent-fair pronunciation assessment, the absence of a deferred-feedback design grounded in Cognitive Load Theory, and the absence of a fused gaze–disfluency correlation metric. A systematic comparison against six current commercial products confirmed that no existing system combines these three capabilities, positioning SpeechMate as a distinct contribution within an increasingly crowded market.

The core loop works end to end, joining a dual-engine router that invokes the code-switch model only when needed, pronunciation scoring that distinguishes regional variation from intelligibility loss, and a gaze–disfluency correlation computed across one timeline. Two observations deserve restating. The differentiating decisions trace to gaps identified in the literature rather than to engineering convenience, which keeps development tethered to a research argument instead of drifting toward feature parity. And the complaint early users raised, unreliable pronunciation scoring which turned out on inspection to concern what the model counts as an error, not how its parameters are tuned: precisely the distinction this paper argues is under-examined.

A. Limitations

Several caveats bound what this paper establishes. The early feedback described above came from a small convenience sample gathered informally, so it motivates the

diagnosis in Section IV.B without validating anything. Everything runs on pretrained inference with no fine-tuning on Malaysian-accented or code-switched data, so the system inherits whatever accent behaviour its checkpoints exhibit; the error rates quoted for the code-switch model are its authors' published figures, not measurements of this system. Code-switch detection is a documented list of roughly 190 terms, not a trained language identifier. GOP scoring, though lighter and more inspectable than an end-to-end neural model, inherits whatever error its forced aligner introduces. The accent-fairness boundary was reasoned from Malaysian-English phonology specifically, so transfer elsewhere needs fresh grounding. The coaching component calls an external LLM, so full offline operation is unsupported, and the educator role weighed during design was dropped.

Two further limits concern measurement and premise. A standard webcam recovers head-and-eye orientation rather than true point of regard, so the tri-zone treatment is deliberately coarse and the correlation runs over an approximation of gaze; it is also computed within one session, over few disfluency events on short recordings, where any correlation is unstable. And deferring feedback costs something: it removes in-session load but forfeits immediate correction, which parts of the feedback literature endorse for well-rehearsed tasks. The claim here is narrower that for anxious speakers on open-ended verbal tasks the load of mid-speech alerts outweighs immediacy and Section III.A argues it from theory rather than establishing it empirically.

B. Future Work

The most direct next step is quantitative evaluation under the pre-registered protocol of Table I and Fig. 6, which requires assembling a Malaysian-accented and code-switched evaluation set with human-verified transcripts. Public Manglish and Malay speech corpora released for research use are suitable for exposing models to accent variation but should not serve as the benchmark: corpora derived from automatically transcribed sources carry label noise that would confound the very measurement the evaluation is intended to produce. Locally recorded, human-verified participant samples are therefore planned as the evaluation ground truth, with the public corpora reserved for adaptation.

Several extensions follow from what is built. A shared timeline means each flagged moment can be replayed as a clip beside its explanation, letting users check why it was flagged rather than trust an opaque aggregate, and stored per-session metrics mean difficulty can track a rolling baseline of the user's own recent sessions, so a first-time anxious speaker is not held to a practised one's standard. Replacing the fixed word list with a trained language identifier would lift the lexicon's coverage ceiling and detect switching at sub-utterance level rather than as a whole-session ratio where a distinction that matters, since a user switching once at a hard word differs from one alternating throughout. Routing itself currently turns on Whisper's language confidence, a proxy rather than a direct measure; a signal trained to predict which engine transcribes a segment more accurately would ground the decision better and make it auditable, which matters because a user routed to the specialised engine presently cannot see why.

Most directly, the deferred-feedback premise rests on theory rather than measurement. A controlled comparison of deferred against real-time feedback, measuring performance

and self-reported anxiety on the same participants, would test it.

Each differentiating mechanism are accent-fair scoring, the fused gaze-disfluency metric, dual-engine routing which rests on a design argument backed by literature and by the absence of any equivalent in existing systems, not on evidence of better user outcomes. What is established is that this combination runs end to end on commodity hardware; what is not established is that any of it helps a speaker improve. Supplying that evidence is the substantive work remaining, and Fig. 6 fixes the thresholds in advance so the answer cannot later be adjusted to suit the system.

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